

Demonym Affix Productivity and Conditioning

As a class, we conducted a survey attempting to measure the productivity of six common demonym affixes. The stimuli were chosen to be real cities which met certain phonological constraints (such as number of syllables) and were not attested in the Corpus of Contemporary American English (COCA) to control against confounding by familiarity. Along with each stimulus, the survey respondents were supplied a continent which may or may not match the actual location of the city in question. In this paper, I explore to what extent geographical location and phonological factors affect the choice of demonym affix.

Initial Exploration

Figure 1 to the right shows the distribution of affixes across the entire dataset. This data suggests that *-ian* and *-an* are the most frequent allomorphs. They are widely applicable and with few restrictions. Next most frequent are *-ese*, *-er*, *-ite*, and null, which are more restricted. The distribution will shift depending on how the data is split over its conditioning factors.

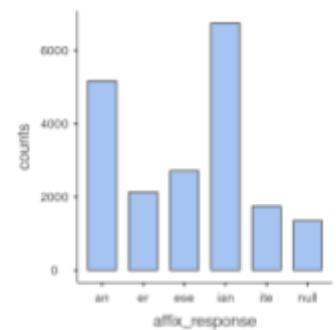


Figure 1: Affix Distribution

Affix Choice by Continent of Question

Part of the exigence of this research project is the claim made in Gordon Danner's article on allomorph competition that the affix *-ese* is primarily used in African and Asian place names (2016), suggesting that affix selection is dependent on geographic location. It would be fitting to start with this question. I conducted a chi-squared test to determine the dependence of affix response on the continent written in the question. The chi-squared value was found to be 727 with 20 degrees of freedom. The p-value was < 0.001 , thus there is likely to be a meaningful relationship here.

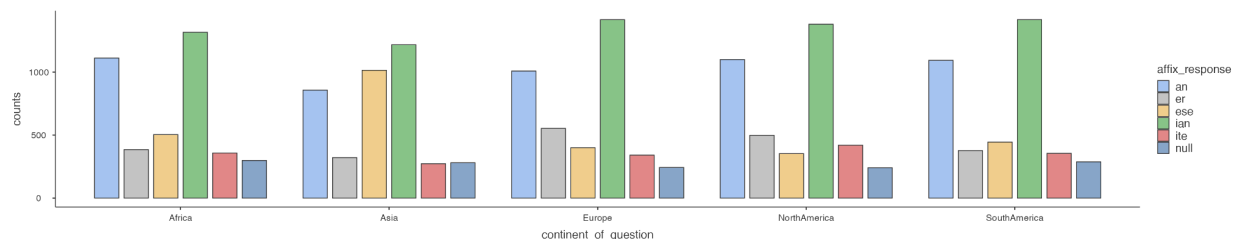
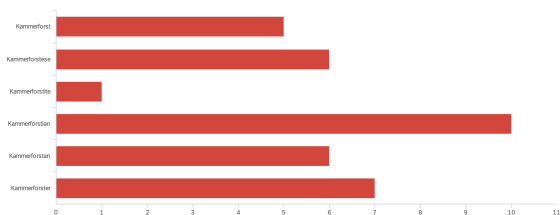


Figure 2: Distribution of Affix Response by Continent of Question

Above, Figure 2 shows the distribution of affix responses by continent. Two trends that stick out include the abundance of *-ese* responses in Asia (as suggested by Gordon Danner) and also the preference for *-er* in Europe. The following charts highlights these two trends:

Question-527 - What would you call a person from Kammerforst (Asia)?



Question-47 - What would you call a person from Kammerforst (Europe)?

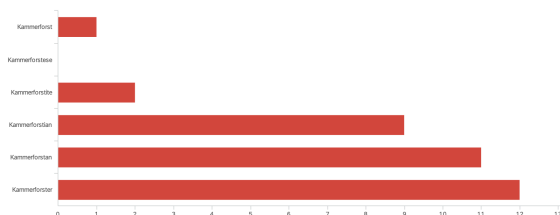


Figure 3: Affix Response Distribution for Kammerforst, Germany

In the case of Kammerforst, placing it in Asia yields a significant number of *-ese* responses. In contrast, placing it in Europe yields zero *-ese* responses, and *-er* secures a majority. This sample shows a strong connection between the geographic location of a certain place and choice of demonym affix.

Affix Choice by Actual Continent

Assuming that a place is in Asia has been shown to favor the *-ese* affix, regardless of whether the place actually exists in Asia. However, this is not the only factor at play. The following two charts both display responses to stimuli that were labeled as Asian, but one shows only places outside of Asia and the other shows only places inside of Asia.

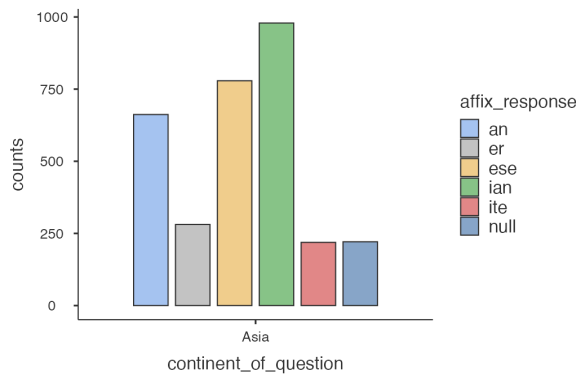


Figure 4: Outside of Asia

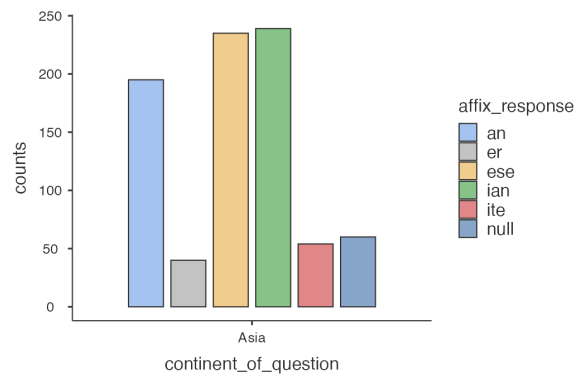


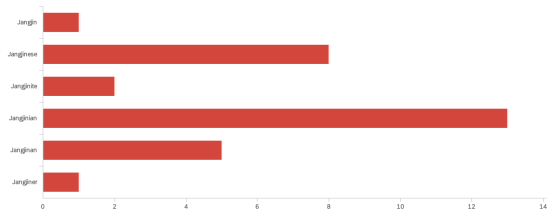
Figure 5: Inside of Asia

Places that were actually inside of Asia show higher proportions of the *-ese* affix than those outside of Asia which were labeled as Asian. Places inside of Asia also notably show lower proportions of *-er*, which is associated more with European places. This means that there must be other qualities about these place names that condition the *-ese* affix besides their geographic association with Asia. Possible explanations will be explored in the sections that follow.

Affix Choice by Etymological Origin

Perhaps a more powerful factor than geographical location is etymological origin, or the language families that it originates from.

Question-154 - What would you call a person from Jangjin (Europe)?



Question-34 - What would you call a person from Jangjin (Asia)?

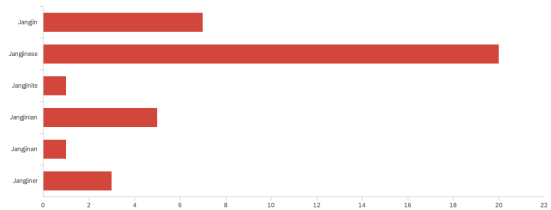


Figure 6: Affix Response Distribution for Jangjin, North Korea

In the case of Jangjin, *-ese* comes out as a frequent response regardless of whether it is placed in Asia or Europe. My hypothesis is that Jangjin's etymological origin is easily recognizable as from Asian language families. A respondent may not be able to tell in particular

that Jangjin is a Korean place name, but they have a sense that Jangjin is unlikely to appear in a European language. Still we see that labeling it as European does draw responses away from the Asian *-ese* and toward the more general *-ian*, although not toward the more European *-er*.

Just like with *-ese*, I hypothesize that the prevalence of *-er* for European places could be due to obvious etymological connections. German and Dutch share the same demonym affix *-er* with English, like with *Berliner* in German and *Nederlander* in Dutch. The English lexicon is also full of Germanic lexemes, so speakers are more easily able to recognize them. This phenomenon is visible elsewhere in the English language where Latinate bases only accept Latinate affixes and Germanic bases only accept Germanic affixes. For example, the Latinate *brutal* accepts the Latinate suffix *-ity* to form *brutality*, but the Germanic *brittle* does not accept *-ity* despite its similar phonological structure.

Affix Choice by Phonological Conditions

Finally, the phonological conditions present at the end of the stem could be more favorable to some demonym affixes than others. I performed another chi-squared statistical test to determine the dependence of affix response on the phonological conditions at the end of the stem (C, CC, or V). The chi-squared value was found to be 760 with 10 degrees of freedom. The p-value was < 0.001 , thus there is likely to be a meaningful relationship here.

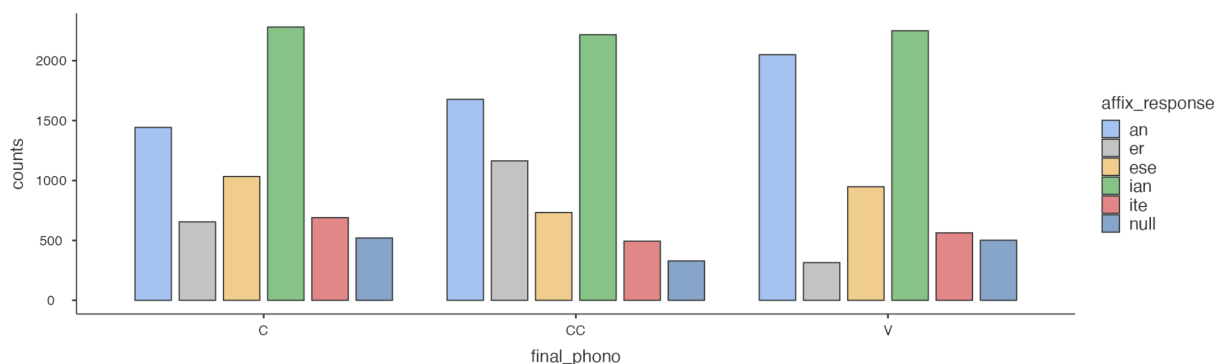


Figure 7: Distribution of Affix Response by Final Phonological Conditions

One notable trend is that *-er* surpasses *-ese* in popularity with final conditions CC. CC endings are common for the European continent. The following tables list the stimuli ending in CC from Asia and South America:

Asia	South America
Oudongk	Andant
Jargalant	Hannaslust
Rashaant	Baamberg
Zakamensk	Estherslust
Keleft	Hoofdkamp
Santmargats	Donderkamp
Fevralsk	Meerzorg
Priargunsk	

Figure 8: Stimuli ending in CC from Asia and South America

The CC ending is very uncommon in the languages of South America, so most of the South American stimuli came from Suriname and are actually of Dutch origin. The CC ending is also very uncommon in the languages of Asia, so most of the Asian stimuli come from Russia and Mongolia, which belong to the Indo-European and Altaic language families respectively. Even though Asia is partial to *-ese*, these examples exhibit a stronger pull to *-er*. This evidence supports my previous hypothesis that etymological origin is a powerful conditioning factor.

Conclusion

Regarding the productivity of demonym affixes, it appears that there are both geographic and phonological conditioning factors at play. Two of the strongest trends identified were the preference for *-er* in Europe and with bases ending in CC and the preference for *-ese* in Asia. Regardless, *-ian* and *-an* are still dominant in most cases as universal demonym affixes.

References

Gordon Danner, Samantha. (2016). Selectional Effects in Allomorph Competition. Proceedings of the Annual Meetings on Phonology. 2. 10.3765/amp.v2i0.3763.